

ETC3460-ETC5346: Week 2 Workshop

Question 1

Using the concepts learned in Week 1 (Slides 1–66), determine whether the S&P 500 index fund would outperform the hedge fund - see Week 1 slide 69

For simple returns, the k-period equity curve is

$$V_t^{(k)} = \prod_{j=0}^{k-1} (1 + R_{t-j})$$

For SP index,

$$\begin{aligned} & (1 + R_{2008}) (1 + R_{2009}) \dots (1 + R_{2015}) \\ = & (1 - 0.37) (1 + 0.266) (1 + 0.151) (1 + 0.021) (1 + 0.16) (1 + 0.323) (1 + 0.136) (1 + 0.014) \\ = & 1.657 \end{aligned}$$

and so \$1 grew to about \$1.657.

For Hedge Funds

$$\begin{aligned} & (1 - 0.239) (1 + 0.159) (1 + 0.085) (1 - 0.019) (1 + 0.065) (1 + 0.118) (1 + 0.056) (1 + 0.017) \\ = & 1.2 \end{aligned}$$

and so \$1 grew to about \$1.20.

The S&P 500 outperformed the hedge fund over 2008–2015.

Question 2

Equity Curve (lecture example). The excel file "Equity_curve_sbux_msft.xlsx" contains monthly prices of Starbucks and Microsoft from Jan 1993 to Dec 2014. Construct the monthly equity curves for both series.